Jiayu Tang

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RESEARCH INTERESTS

I am interested in utilizing interdisciplinary approaches to study how antibodies bind antigens. I aim to use AI rigorously, grounded in and validated against experimental data, to build a mechanistic understanding of antibody–antigen interactions that can inform clinical and industrial decisions as well as frontier research.

EDUCATION

M.S., Food Science and Technology; Minor in Statistics 08/2024 - 08/2026
University of Nebraska-Lincoln, Lincoln, NE, USA GPA 3.95/4.00

B.Sc., Food Science and Technology (Dual Degree) 09/2020 - 06/2024
University of Nebraska-Lincoln, Lincoln, NE, USA GPA 3.52/4.00

B.Eng., Food Science and Engineering (Dual Degree)
Northwest A&F University, China

RESEARCH EXPERIENCE

OPS Research Assistant 10/2026 - Present
Department of Infectious Diseases and Immunology, College of Veterinary Medicine, University of Florida
Supervisor: Dr. Yiquan Wang

- Develop in-silico antibody-design pipelines built on framework grafting, native-structure controls, and epitope-masked structural evaluation.
- Run deep-learning model (AlphaFold3, ESMFold2) and antibody design workflows (RFAntibody, BindCraft2, RFD4) for antibody-antigen predictions and evaluate model confidence, docking accuracy, and interface contacts.
- Use AI coding agents (Claude Code, Codex) to plan, implement, and audit computational workflows.

Graduate Research Assistant 07/2024 - 07/2026
Department of Food Science and Technology, University of Nebraska-Lincoln
Advisor: Dr. Edward Deehan

Clinical trials:

- Coordinated a 12-week, three-phase randomized crossover trial of dry bean market-class effects on gut microbiota and cardiometabolic health markers (IRB Project #: UNL-00024307; ClinicalTrials.gov #: NCT06935435).
- Co-developed the NIH-style study protocol, informed consent form, and recruitment materials; authored 4 standard operating procedures (SOPs) for Deehan Lab.
- Ran 200+ clinic visits and oversaw recruitment, scheduling, documentation, and study supplies.
- Assisted a human biobank study that collect human blood and stool samples with health information (IRB Project #: UNL-00023846).
- Collected and analyzed clinical data in R and Python with multivariate and regression methods.

Systematic review:

- Led a systematic review project to evaluate the effects of dietary pulses market classes on gut microbiome features and health-related outcomes across peer-reviewed acute and long-term intervention clinical trials (PROSPERO Registration #: CRD420261350487).
- Searched databases and screened 1,800+ records with machine-learning tools (ASReview, TERA-Tool).
- Applied a large language model (Gemini 3 Pro) to assist data extraction, with every extracted value verified by two human reviewers.

Undergraduate Research Assistant 04/2022 - 06/2023

College of Food Science and Engineering, Northwest A&F University
Advisor: Dr. Chunxia Xiao

- Co-led an *in vitro* fermentation project testing how three lactic acid bacteria species change the phenolic content, antioxidant activity, and starch digestibility of *proso* millet (Undergraduate Thesis).
- Measured total phenolic content (Folin–Ciocalteu, gallic acid equivalents) and DPPH and ABTS radical-scavenging activity by spectrophotometry (co-author, Foods 2023).
- Synthesized findings from 50+ articles on coarse grain consumption and type 2 diabetes for a review article (co-author, J Sci Food Agric 2023).

PUBLICATIONS

*[#] co-first author; * corresponding author*

Manuscript in preparation

1. **Tang J**[#], Adinchezor C[#], Grimaldo C, Zhang Y, Iwuamadi I, Rasmussen C, Rose D, Majumder K, Deehan E*. Exploring the impact of different dietary pulse market classes on human gut microbiota and health.
2. **Tang J**[#], Rasmussen C[#], Deehan E*. Effect of distinct dietary pulse market classes on gut microbiome and health markers: a systematic review of intervention trials.
3. Grimaldo C, **Tang J**, Whisenhunt L, Chai W, Deehan E*. Evaluating interrelationships between body composition and immunometabolic dysfunction using bioelectrical impedance analysis: an observational study.

Peer-reviewed

1. Xiao J, Li Y, Niu L, Chen R, **Tang J**, Tong Z, Xiao C*. Effect of adding fermented proso millet bran dietary fiber on micro-structural, physicochemical, and digestive properties of gluten-free proso millet-based dough and cake. Foods. 2023;12(15):2964.
2. Deng X, Niu L, Xiao J, Guo Q, Liang J, **Tang J**, Liu X, Xiao C*. Involvement of intestinal flora and miRNA into the mechanism of coarse grains improving type 2 diabetes: an overview. J Sci Food Agric. 2023;103(9):4257-4267.

AWARDS

Excellence in the Department of Food Science and Technology 2025
Department of Food Science and Technology, University of Nebraska-Lincoln

Mary and Charles C. Cooper / Emma I. Sharpless Fellowship 2024
College of Agricultural Sciences and Natural Resources, University of Nebraska-Lincoln

Undergraduate Merit Scholarship (3 consecutive years) 2020 – 2023
College of Food Science and Engineering, Northwest A&F University, China

FUNDING & GRANT WRITING

Chancellor's Graduate Research Grant 2026
$400, University of Nebraska-Lincoln

Graduate Research Assistantship 2024 – 2026
(Full stipend; tuition remission), Department of Food Science and Technology, University of Nebraska-Lincoln

TALKS & PRESENTATIONS

Tang J, “Exploring the impact of different dietary pulse market classes on human gut microbiota and health” 2026

Nebraska Food Innovation Center, University of Nebraska-Lincoln (presentation)

Rasmussen C (Mentee), **Tang J**, et al., “Effect of different Nebraska dry bean market classes on human health and the gut microbiota: a preliminary analysis” 2026

Nebraska Research Days, University of Nebraska-Lincoln (poster)

Rasmussen C (Mentee), **Tang J**, et al., “Dietary pulse market class-specific effects on gut microbiota, metabolites, and human health: a systematic review” 2026

Nebraska Research Days, University of Nebraska-Lincoln (poster)

Tang J, et al., “Effect of different Nebraska dry bean market classes on human health and the gut microbiota: a 3-phase randomized crossover clinical trial” 2026

Missouri Valley Branch ASM Annual Meeting (poster)

Tang J, et al., “Exploring the impact of different dietary pulse market classes on host metabolic health and gut microbiome” 2025

Food Science and Technology Symposium, Nebraska Research Days, and NPOD 16th Annual Spring Retreat, University of Nebraska-Lincoln (poster)

TEACHING EXPERIENCE

Teaching Assistant, FDST 458, FDST 458-153 2026
University of Nebraska-Lincoln, Lincoln, NE
- Led lab sessions and graded coursework.

Guest Lecturer, FDST 498 2025
University of Nebraska-Lincoln, Lincoln, NE
- Gave three lectures on systematic review methods and AI-assisted screening.

Teaching Assistant, FDST 412, FDST 429, NUTR 250 2023 - 2024
University of Nebraska-Lincoln, Lincoln, NE
- Graded assignments and exams.

Teaching Assistant (Internship) 2022 - 2023
Xi’an New Oriental Education & Technology Group, China
- Coached on IELTS exam preparation.

Teaching Assistant, COMM 209 2022
Northwest A&F University, China
- Graded assignments and exams.

LEADERSHIP & MENTORSHIP

Leadership

Graduate Representative 2025 - 2026
Graduate Student Association, University of Nebraska-Lincoln, Lincoln, NE
- Advocated for food science graduate students on academic affairs.

Graduate Student Advisory Board Member 2024 - 2025
University of Nebraska-Lincoln, Lincoln, NE
- Founded a peer-mentoring program pairing new graduate students with mentors.

Graduate Representative 2024 - 2025
College of Agricultural Sciences and Natural Resources, University of Nebraska-Lincoln, Lincoln, NE
- Organized three professional development workshops for graduate students.

Communications Lead 2020 - 2022
Northwest A&F University, China
- Directed a 13-member team producing weekly newsletters and videos.

Mentorship

Cade Rasmussen, Mentor & Co-supervisor, *University of Nebraska-Lincoln* 2025 – 2026

Luo Lin, Mentor, *University of Nebraska-Lincoln* 2025 - 2026
Yaxian Yang, Mentor, *University of Nebraska-Lincoln* 2025

SERVICE

Volunteer, *Nebraska Food for Health Center* 2025
Junior Reviewer, *Clinical Nutrition* 2024
Academic Advisor, *Northwest A&F University* 2022-2023
Academic Coordinator, *College of Food Science and Engineering, Northwest A&F University* 2021-2023
Academic Affairs Assistant, *Office of Undergraduate Studies, Northwest A&F University* 2022

PROFESSIONAL DEVELOPMENT, AFFILIATIONS & MEMBERSHIPS

Certifications

Python for Data Science, AI & Development, *Coursera* 2025
Data Science: R Basics, *HarvardX* 2025
Alberta Food Safety Basics, *University of Alberta Health Service* 2025
GCP for Clinical Trials with Investigational Drugs and Biologics, *CITI Program* 2024
Biomedical Responsible Conduct of Research, *CITI Program* 2024
Conflicts of Interest, *CITI Program* 2023
Biomedical Investigators and Key Personnel, *CITI Program* 2023

Workshops & training

High-Performance Computing for Researchers, *University of Nebraska-Lincoln* 2025
Exploring Your Personal Approach to Leadership & Collaboration, *Clifton Strengths Institute* 2025
Grant Writing Seminar, *University of Nebraska-Lincoln* 2025
Undergraduate Research Experience (FDST 498), *University of Nebraska-Lincoln* 2024

Affiliations & memberships

American Society for Microbiology (ASM) 2024 - 2026
American Society for Nutrition (ASN) 2024 - 2026
International Scientific Association for Probiotics and Prebiotics (ISAPP) 2023 - 2025

SKILLS

Computational antibody modeling: AlphaFold structure prediction; *de novo* antibody-design pipelines (RFantibody, BindCraft2); AI coding agents (Claude Code, Codex) for building and validating workflows
Statistics: experimental design and power analysis; multiple regression and multivariate analysis; interpretation in biological and clinical context
Wet lab: PCR; human blood and stool sample processing; in vitro fermentation; spectrophotometric assays
Clinical research: propose, design, and lead human clinical trials; protocol and SOP development; IRB/GCP procedures; ClinicalTrials.gov registration; participant recruitment and visit coordination; grant proposal writing
Evidence synthesis & communication: systematic review; LLM-assisted data extraction with human validation; grant writing
Languages: Mandarin (native), English (fluent)